Yanfei Zhu

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EDUCATION

University of Wisconsin-Madison **Madison, WI**
VISP, M.A. in Mathematics, GPA: 3.844/4.000 *Sep. 2024 – May 2026*
Visiting International Student Program (VISP)

- **Courses:** Introductory Topology (1&2, Algebraic Topology), Homological Algebra, Complex Analysis, Abstract Algebra, Ordinary Differential Equations, Modern Algebra, Analysis II

Southern University of Science and Technology (SUSTech) **Shenzhen, CN**
B.S. in Mathematics and Applied Mathematics *Sep. 2021 – Jun. 2025*

- **Weighted Average Score:** 88.09/100
- **Courses:** Algebraic Topology, Topology, Representation of Groups, Abstract Algebra, Advanced Linear Algebra, Real Analysis, Complex Analysis, Mathematical Analysis, Partial Differential Equations, Ordinary Differential Equations, Discrete Mathematics, Elementary Number Theory, Probability and Statistics, Calculus

Research Interests: I am interested in using algebraic and topological methods to study geometric problems, especially those involving manifolds. My current research focuses on equivariant homotopy theory and homological stability. I am also interested in optimization on manifolds.

RESEARCH AND READING EXPERIENCE

Directed Reading in Manifold Optimization *Jun. 2026 – Present*
Advisor: Prof. Qinglin Tang, Sichuan University
Topics: Manifold optimization

Directed Reading in Intersection Homology *Aug. 2025 – Dec. 2025*
Advisor: Prof. Laurentiu Maxim, Mathematics Department, UW–Madison
Topics: Intersection homology, sheaves on topological spaces, Verdier duality

Directed Reading in Riemannian Geometry and Morse Theory *Mar. 2025 – Oct. 2025*
Advisor: Prof. Ruobing Zhang, Mathematics Department, UC San Diego
Topics: Riemannian geometry (Petersen, do Carmo), Morse theory (Milnor)

Directed Reading in Homological Stability *Apr. 2025 – Jul. 2025*
Advisor: Prof. Yifei Zhu, Mathematics Department, SUSTech
Topics: Homological stability, configuration spaces, FI-modules, braid groups

Undergraduate Thesis *Dec. 2024 – Jul. 2025*
Symmetry of Solutions in Nonlinear Aggregation-Diffusion Equations and Its Applications
Advisor: Prof. Linlin Su, SUSTech
Studied symmetry properties of stationary solutions to nonlinear aggregation-diffusion equations, with a focus on radial symmetry and applications to numerical methods.

Reading Seminar in Homological Algebra *Sep. 2023 – Dec. 2023*
Advisor: Prof. Ziqing Xiang, Mathematics Department, SUSTech
Topics: Category theory, homological algebra

Details and notes for selected seminar and reading presentations are available on my personal website.

Preprints

The RO(C_q)-Graded Cohomology of Complete Flags in the Regular Representation
Preprint, 2026.
Computes the genuine equivariant cohomology ring, Schubert products, and fixed-point restriction for complete flags in the complex regular representation of C_q, for odd primes q.

Two Lifted Actions on Oriented Configuration Spaces: Stabilization and Bredon Homology
Preprint, 2026.
Finite generation of integral Bredon homology for oriented configuration spaces, using both lifts of a reflection action.

Full texts of the preprints are available on my personal website.

TEACHING EXPERIENCE

Grader, Department of Mathematics, UW–Madison *Feb. 2026 – May 2026*
Graded undergraduate mathematics assignments and provided written feedback.

SUMMER SCHOOLS AND WORKSHOPS

IWoAT Summer School, Chinese Academy of Sciences *Aug. 2026*
Beijing, CN. Topics: Stable homotopy theory and brave new algebra.

IWoAT Summer School, Westlake University *Jul. 2025*
Hangzhou, CN. Topics: Adams spectral sequence computations; the last Kervaire invariant problem.

Summer School, University of Science and Technology of China *Jul. 2025*
Hefei, CN. Topics: Geometry and analysis.

SKILLS AND LANGUAGES

Technical Skills: LaTeX, Python, MATLAB, Java
Languages: Mandarin (native), English (fluent), Japanese (basic)